

Transformación rápida de Fourier

Sean

- $N = 2^n$ con n entero positivo.
- $N = 2M$ con M entero positivo.
- $W_N^n = e^{\frac{-j2\pi n}{N}}$.

$$\begin{aligned} F(u) &= \frac{1}{2M} \sum_{x=0}^{2M-1} f(x) e^{\frac{-j2\pi ux}{2M}} = \frac{1}{2M} \sum_{x=0}^{2M-1} f(x) W_{2M}^{ux} \\ &= \frac{1}{2} \left[\frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_{2M}^{u(2x)} + \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_{2M}^{u(2x+1)} \right] \end{aligned}$$

Donde

$$\begin{aligned} W_{2M}^{u(2x)} &= e^{\frac{-j2\pi u(2x)}{2M}} = e^{\frac{-j2\pi ux}{M}} = W_M^{ux} \\ W_{2M}^{u(2x+1)} &= e^{\frac{-j2\pi u(2x+1)}{2M}} = e^{\frac{-j2\pi u(2x)}{2M}} e^{\frac{-j2\pi u}{2M}} = W_{2M}^{2ux} W_{2M}^u = W_M^{ux} W_{2M}^u \end{aligned}$$

reemplazando se tiene

$$F(u) = \frac{1}{2} \left[\frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_M^{ux} + \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_M^{ux} W_{2M}^u \right]$$

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$$\begin{aligned} F_{even}(u) &= \frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_M^{ux} \\ F_{odd}(u) &= \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_M^{ux} \end{aligned}$$

$F(u)$ se reduce a

$$F(u) = \frac{1}{2} [F_{even}(u) + F_{odd}(u) W_{2M}^u]$$

Por otro lado de

$$W_M^{u+M} = e^{\frac{-j2\pi(u+M)}{M}} = e^{\frac{-j2\pi u}{M}} e^{\frac{-j2\pi M}{M}} = e^{\frac{-j2\pi u}{M}} e^{-j2\pi} = W_M^u$$

$$W_{2M}^{u+M} = e^{\frac{-j2\pi(u+M)}{2M}} = e^{\frac{-j2\pi u}{2M}} e^{\frac{-j2\pi M}{2M}} = e^{\frac{-j2\pi u}{2M}} e^{-j\pi} = -W_{2M}^u$$

se observa

$$\begin{aligned}
F(u+M) &= \frac{1}{2M} \sum_{x=0}^{2M-1} f(x) W_{2M}^{(u+M)x} \\
&= \frac{1}{2} \left[\frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_M^{(u+M)x} + \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_M^{(u+M)x} W_{2M}^{u+M} \right] \\
&= \frac{1}{2} \left[\frac{1}{M} \sum_{x=0}^{M-1} f(2x) W_M^{ux} + \frac{1}{M} \sum_{x=0}^{M-1} f(2x+1) W_M^{ux} (-W_{2M}^u) \right] \\
&= \frac{1}{2} [F_{even}(u) - F_{odd}(u) W_{2M}^u]
\end{aligned}$$